

LunarAI — Final Judges Report

SIH 26166 · 2026-09-27 19:49:33 · 83 validation runs

Evaluation Metrics

retrieval_top1	0.7500
retrieval_top5	0.9830
retrieval_mAP	0.7900
inlier_ratio_median	0.9820
coverage_median	0.6720
rmse_gt_before_ecc_median_px	0.5247
rmse_gt_after_ecc_median_px	0.2450
subpixel_share	0.9210
patches_indexed	940
lunadna_checkpoint_epoch	1

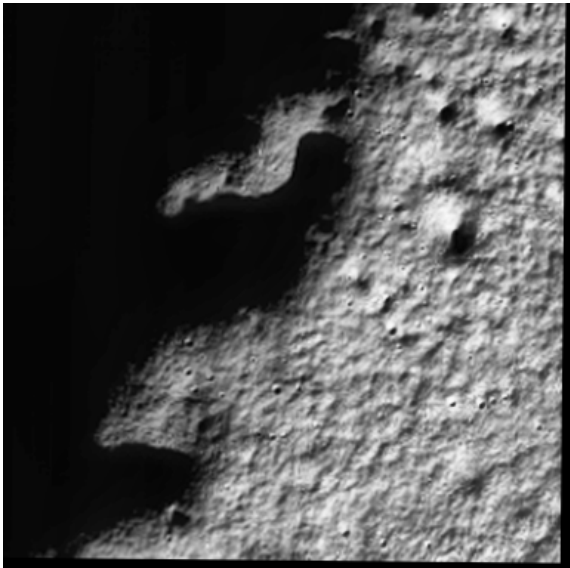
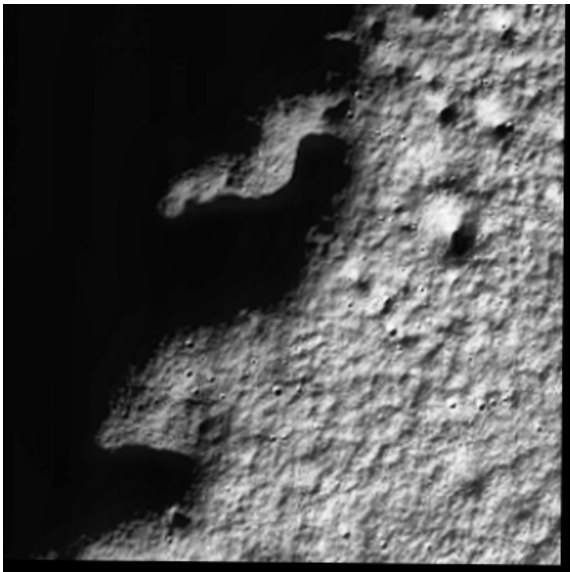
Summary

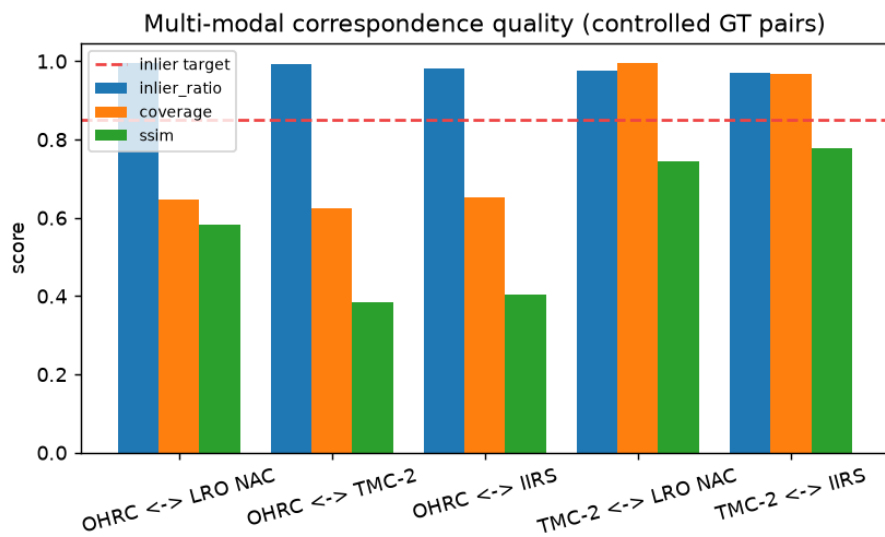
Why LunaDNA?: A plain matcher cannot know *where* on the Moon a patch is from. LunaDNA's 512-D terrain fingerprint

Why FAISS before registration?: Matching the whole lunar database is unbounded work; retrieval narrows it to 1

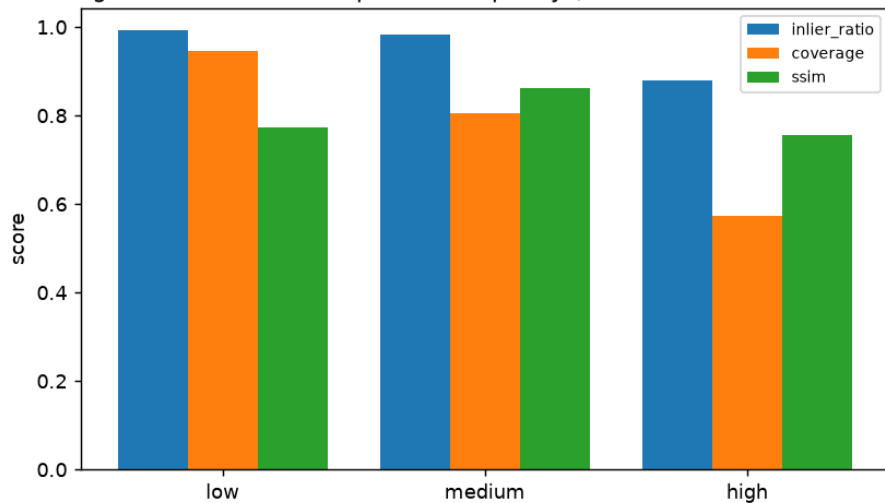
How does LunarAI handle scale variation?: At two levels. (1) GSD: the pipeline resamples every product to a co

How does LunarAI handle sun-angle variation?: CLAHE normalization at preprocessing (applied to both images bef

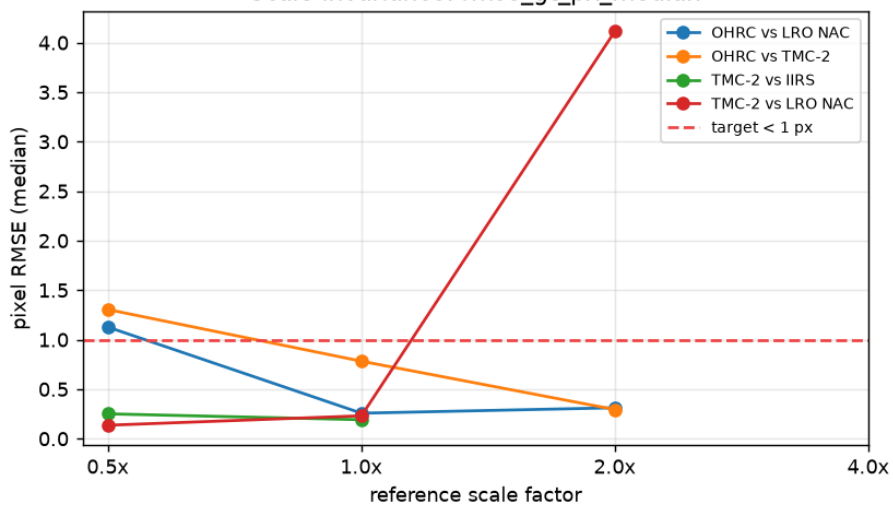




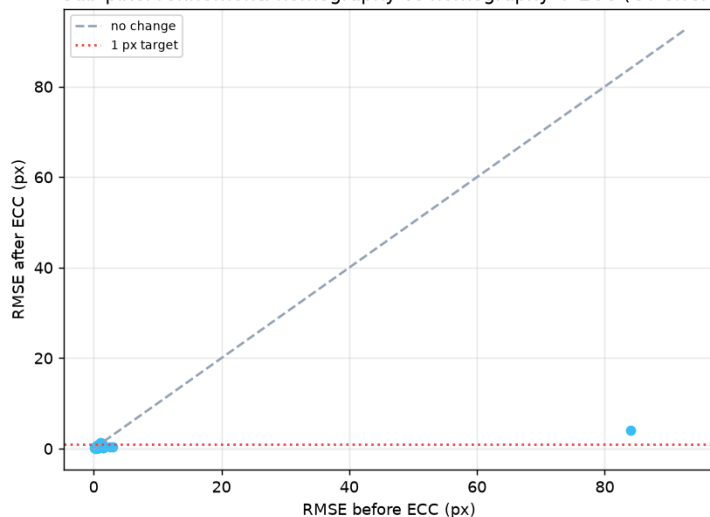
Sun-angle robustness: correspondence quality (controlled illumination tran:



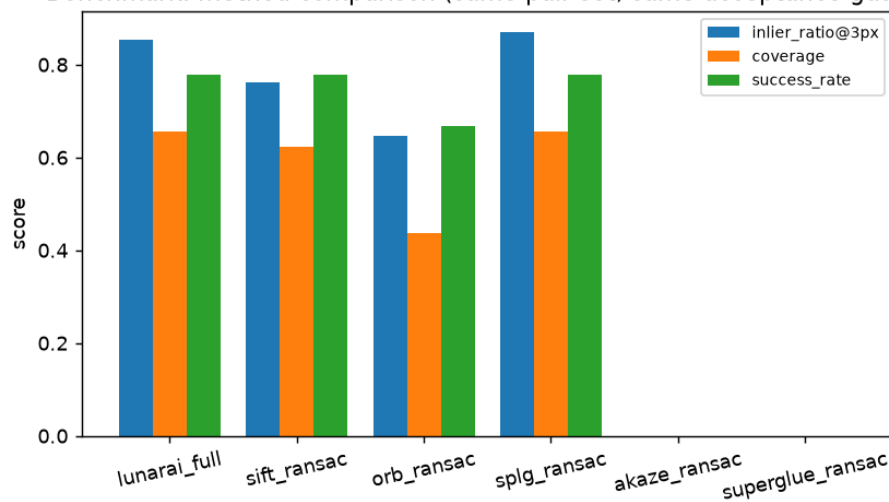
Scale invariance: rmse_gt_px_median



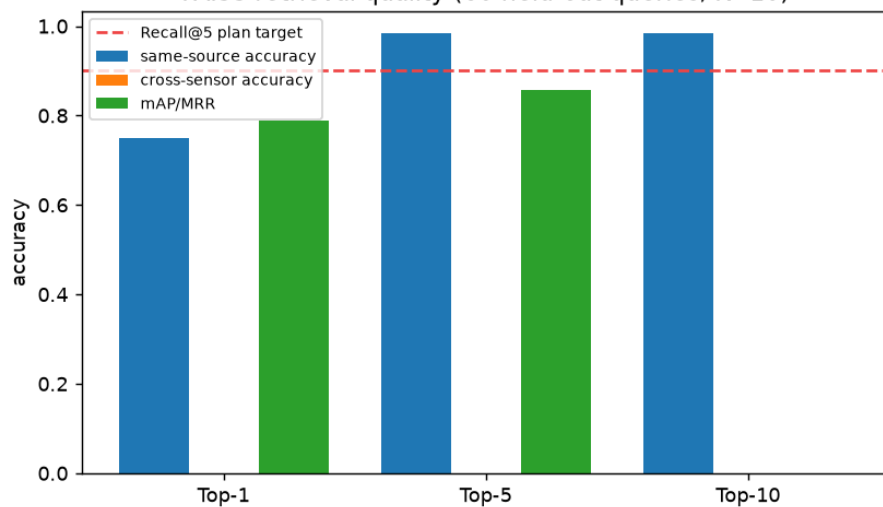
Sub-pixel refinement: homography vs homography + ECC (GT error)



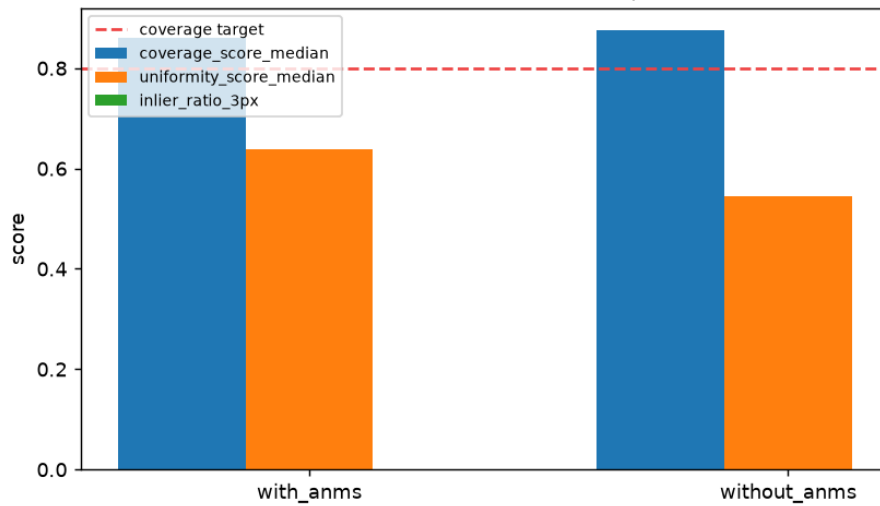
Benchmark: method comparison (same pair set, same acceptance guard)



FAISS retrieval quality (60 held-out queries, K=10)



Uniform match distribution: without vs with ANMS (median over solved pa



Ablation study: component contribution (residual inliers @3px)

